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Personalized Lifestyle Modifications for Improved Metabolic Health: The Role of Cultural Sensitivity and Health Communication in Type 2 Diabetes Management

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ABSTRACT

Type 2 Diabetes (T2D) is a chronic multifaceted metabolic condition characterized by elevated blood glucose levels with varying degrees of insulin resistance and abnormal insulin production. Lifestyle modifications, such as those defined by the guidelines for diabetes self-management education and support (DSMES), are foundational for glycemic control. A current gap in T2D management is addressing DSMES which is tailored to best serve the diversity of patients with this disease. The purpose of this narrative review is to discuss the current literature related to lifestyle modification for T2D, the importance of culturally sensitive T2D management programs, and the impact culturally sensitive and diverse T2D management programs have on cardiometabolic health. Despite being disproportionately affected by T2D, racial and ethnic minorities have low referral rates for DSMES. This growing disparity may be exacerbated by a lack of awareness of how to adapt lifestyle modifications in a culturally competent manner and how social determinants of health (SDOH) may affect the infrastructure and resources available to diverse patient populations. Currently, there is limited research on how DSMES is addressed to include culturally and literacy-sensitive recommendations in the United States. Patient-centered care, emphasizing personalized DSMES, can lead to improved glycemic control, reduced healthcare costs, and improved cardiometabolic health. A key component of effective DSMES should include culturally and literacy-sensitive approaches with an awareness of the impact of SDOH. Understanding how ethnicity, race, and culture influence experiences with T2D management can help providers prescribe more patient-centered and sustainable recommendations.

KEY TEACHING POINTS

1. A key component of Type 2 Diabetes (T2D) management is a combination of lifestyle and behavioral modifications, such as those defined by the guidelines for diabetes self-management education and support (DSMES).
2. Patient-centered care with an emphasis on DSMES may result in improved and sustained glycemic control, reduced healthcare costs, and improved quality of life; however, racial/ethnic minority groups, who already share a disproportionate burden of diabetes risk, diagnosis, and complications, experience proportionally low rates of referral for DSMES.
3. Cultural nuances do, and should, influence DSMES recommendations. Contextualizing and addressing the unique cultural needs and preferences for individual diabetes patients is vital for promoting optimal care and outcomes.
4. Diabetes providers should understand how social determinants of health (SDOH) and factors such as context, community, and communication, may affect the infrastructure and resources available to diverse patient populations for self-management.
5. Patient-centered diabetes care that appreciates the influence of cultural and SDOH factors may result in more effective DSMES, and providers should be aware of when and how to adapt DSMES for diverse patients leveraging cultural competency and humility skills at the point of care.

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Introduction

Type 2 Diabetes (T2D) is a chronic multifaceted metabolic condition characterized by persistently elevated blood glucose levels and varying degrees of insulin resistance and abnormal insulin production. T2D affects over 34.2 million people in the United States (US) with another estimated 88 million individuals at increased risk of developing diabetes (i.e., pre-diabetes)(1). Much research has been conducted to understand the factors that contribute to diabetes risk and poor outcomes. Current evidence suggests these contributing factors may stem from multiple factors including genetic, epigenetic, and environmental sources (2). Poorly controlled T2D may lead to significant systemic complications such as an increased risk for cardiovascular, neurological, renal, skin, muscular, and ocular-related disorders (3). T2D also has a widespread impact that extends into psychological, social, and even financial wellbeing (4). As of 2017, the US spends over \$237 billion on diabetes care, which includes physician visits, prescription medications, hospital inpatient care, and missed work or reduced productivity costs. In total, it is estimated that the health care cost for an American with T2D is 2.3x higher than for an individual without diabetes (1).

In light of the high cost and prevalence of T2D and its complications, much effort over the last decade has focused on the development of novel therapeutics for the management of T2D (5). Lifestyle and behavioral modifications, such as those defined by the guidelines for diabetes self-management education and support (DSMES), however, remain the bedrock and most cost-effective option for prevention and a vital component of management for T2D (6,7). The American Diabetes Association (ADA) and the Association of Diabetes Care & Education Specialists developed the national DSMES standards which are updated every 5 years in the US (8). They define DSMES as a process that facilitates the acquisition of behaviors and skills needed to effectively manage diabetes and the support systems required to sustain these behaviors throughout the lifespan of living with diabetes. Principle components of effective DSMES focus on empowering people with diabetes to self-manage through informed, collaborative decision-making alongside their healthcare team, implementing care plans, problem-solving, and navigating diabetes distress. Hallmarks of the requisite skill set for effective diabetes self-management include dietary knowledge and behaviors, personalized physical activity prescription, medication adherence, glucose monitoring & technology use, coping & troubleshooting strategies, and risk reduction efforts (Figure 1). DSMES is often paired with targeted medical nutrition therapy (MNT), and increasingly recognizes the need to address cultural factors and utilize person-centered approaches in its implementation. A review of DSMES in the United Kingdom found that under-represented minority patient populations were amongst the individuals who most benefited from DSMES programs. Successful programs had several themes in common, including a focus on structured, repeated educational sessions conducted in both group and individual settings, in person, and delivered by a trained and qualified facilitator that could personalize the information to each patient (9). Used either as a standalone approach or in

combination with pharmacotherapies, it is well accepted that DSMES yields systemic benefits and help regulate broader metabolic risks beyond glycemic control including improved cholesterol and triglycerides profile, body composition, and body mass index (BMI) (10). However, DSMES is most effective when it is personalized to each patient in order to maximize its role in diabetes management (9).

Patient-centered care with an emphasis on DSMES leads to improved and sustained glycemic control, reduced health-care costs, and improved quality of life (7,11,12). Unfortunately, many racial/ethnic minority groups who already share a disproportionate burden of diabetes risk, diagnosis, and complications also experience proportionally low rates of referral for DSMES (13) despite the elevated risk of poor diabetes control compared to Non-Hispanic White (NHW) populations. Furthermore, evidence from the Diabetes Prevention Program demonstrated that lifestyle intervention had a comparable impact on diabetes risk across racial/ethnic groups, suggesting that tailored approaches to address lifestyle may also be beneficial in the management of T2DM (14). It is essential that both providers and patients understand and have access to proper DSMES resources, and that discussions around lifestyle and behavioral adjustments account for cultural and sociodemographic factors that may impede the patients' capacity to effectively incorporate these recommendations into their diabetes management.

A current gap in T2D management is addressing DSMES tailored to best serve the diversity of patients with this disease (15,16) in the US. Lifestyle modifications are often prescribed in a general sense but may not be considered, or adapted, to diverse patient populations. This is a key component of diabetes education (16) as T2D disproportionately affects racial and ethnic minorities, particularly African American, Hispanic, Southeast Asian, and Native Americans (17). Ensuring that providers have the knowledge base, support system, and guidance on up-to-date DSMES recommendations for T2D that can be adapted, contextualized, and personalized for an increasingly diversified patient population is a keystone of patient-centered care, and has been shown to lead to improved effectiveness and health outcomes (18). To date, there is minimal research on how DSMES is addressed to include culturally sensitive and patient-centered recommendations in the US and even less about what systems are in place to help bridge the gap between the recommendations and patient-personalization. The purpose of this narrative review is to critically assess the current landscape of the DSMES-based literature, discuss the importance of culturally sensitive T2D management programs, and explore the impact culturally sensitive and diverse T2D management programs may have on overall health and glucose-related outcomes in other settings that would translate positively to the broader US landscape.

DSMES, nutrition & physical activity, and cultural inclusivity

Glucose metabolism is a tightly regulated biological process responsible for ensuring glucose is readily available in



Figure 1. Diabetes Self-Management Education & Support (DSMES) Hallmarks: DSMES focuses on empowering people with diabetes to self-manage through several factors. Hallmarks of the requisite skills needed to employ these factors include dietary knowledge and behaviors, personalized physical activity prescription, medication adherence, glucose monitoring & technology use, coping & troubleshooting strategies, and risk reduction efforts.

sufficient quantities for vital organs. Although managed internally, glucose metabolism is also particularly sensitive to external factors, such as dietary profile (i.e., calorie intake, calorie quality, and total carbohydrate level intake) and physical activity profile (i.e., exercise type, daily energy expenditure level, skeletal muscle activation, and glucose uptake/usage). Nutrition and exercise have direct and distinct impacts on glucose metabolism and both lead to benefits that extend into other aspects of metabolic health. It is well accepted that improved nutritional habits combined with physical activity increase skeletal muscle mass tissue quality and quantity, which leads to sustained, systemic glycemic control as well as improved insulin sensitivity, glucose uptake, blood pressure management, and body composition (19). However, the approaches and available literature exploring how to design and prescribe personalized and culturally

inclusive programs are lacking and may impact how nutrition and exercise programs translate to diverse populations.

A key component of patient-centered diabetes care is identifying which dietary approach best fits the individual including flexibility, medication program, and cultural and sociodemographic considerations. Currently, the federal dietary guidelines lead the efforts on addressing and improving dietary prescription to maintain and improve health and indirectly impact diabetes diet prescription. Reports highlight the implicit bias in the federal dietary guidelines, where much of the focus is primarily on Caucasian/White populations (20,21). For example, there is an emphasis on dairy products for their rich mineral content and health additions in a well-designed diet. Yet, lactose intolerance and sensitivity are common amongst several diverse groups. This has led to a call for more inclusive federal guidelines, where as

recent as 2020, the Physicians Committee in the U.S. had penned a letter to the United States Department of Agriculture (USDA) and the Department of Health and Human Services (DHHS), to widen the federal dietary guidelines for more inclusive food options (22). Despite this, the federal dietary guidelines have not been developed with a cultural inclusivity mindset. This is observed in several aspects of dietary guidance, such as suggested foods to consume, meal programming, the BMI guidelines used as part of diagnostic and progress tracking, and more.

In 2019, the American Diabetes Association (ADA) dietary recommendations took a generalized stance that focused on promoting and supporting “healthful eating patterns”. The consensus report emphasized consuming a variety of nutrient-dense foods in appropriate portion sizes to improve overall health and “1) achieve and maintain body weight goals 2) attain individualized glycemic, blood pressure, and lipid goals, and 3) delay and prevent complications of diabetes” (23). Current diet recommendations from the ADA emphasize personalization rather than a strict, uniform prescription that encourages the addition of non-starchy vegetables, minimizing added sugars and refined grains, and an increase in whole foods and minimally processed foods over processed and poor-quality foods (24).

This openness for more personalized prescriptions lends itself to creating research studies and programs that incorporate more diverse and inclusive approaches. Results from these studies could improve research design practices and aid in areas such as recruitment, retention, and more impactful results. For example, Hurt et al. (25) led a 2-part pilot study designed to include bidirectional communication and focus groups with Black men in the US to gain insight into how to better develop gender and culturally-inclusive Type 2 diabetes education programs that lead to healthier outcomes in African American men (25). Part 1 focused on types of diabetes prevention programs the men would be interested in participating and Part 2 focused on diet and physical activity preferences. The participants provided feedback on recipe taste-testing, appealing foods, and texture preferences. The research team reflected that recipes were adjusted to best fit the nutritional needs *as well as* the participants’ preferences. They also discussed some of the barriers related to healthy goals such as busy schedules and lack of resources. This information helped inform the research team how to better develop larger studies and aided in more personalized programming for practitioners and providers (25).

Another example of how a bidirectional research model can provide valuable insight is observed through the study conducted by Prinjha et al. (26). This team designed a research study on the “seldom heard” views of British South Asian people with Type 2 Diabetes. Through focus groups, they identified that cultural competencies (amongst other considerations) were essential components of developing more personalized, dietary guidance for diabetes management (26). Topics such as religious considerations, challenges to adhering to a healthy diet, and thoughts on what a healthy diet would look like were discussed. Similarly, Hempler *et al.* engaged with an immigrant population in Denmark to “co-create” a DSMES program that was tailored

to the needs of the target population in the domains of diabetes knowledge, behaviors, and mental health support with active and iterative feedback from the target group prior to implementation. They identified short-term improvements in several important diabetes-related outcomes including A1C, self-reported wellbeing, and dietary behaviors which they attributed to the enhanced cultural sensitivity of the intervention gained through the co-creation process (27). While these studies are among a growing number focused on diverse populations, more funding and opportunities are needed to continue to enhance this much-needed area of research and address the major gaps that remain in the literature. This includes which culturally-sensitive approaches may serve as best practices and have potential for the greatest long-term impact on outcomes.

The best T2D management programs include a prescribed aspect of regular physical activity. Physical activity supports improved glucose management through several distinct mechanisms such as increased glucose uptake by the contracting skeletal muscle, increased insulin sensitivity, decreased fasting plasma insulin, and reduced hepatic glucose production (28,29). In some cases, exercise may reduce reliance on pharmacological agents and reduce medications and associated costs (30). Like nutrition, exercise induces a multiplicity of pathways that cumulatively impact T2D management including improved gut microbiome (31), cardiovascular health and fitness (32,33), cell membrane integrity, oxidative stress management pathways (34), psychological stress management (35), appetite (36), hormone signaling (37), and restful sleep (38,39).

Despite the systemic, positive phenotypic outcomes associated with regular physical activity, the 2020 National Health Interview Survey reported that only 24.2% of adults met the 2018 physical activity recommendations. Physical activity participation is impacted by several factors, including cultural background, socioeconomic status, and education level (40). Currently, The American College of Sports Medicine (ACSM) and the ADA recommend 150 min of moderate to vigorous-intensity activity spread out over three or more days each week. Moderate to vigorous-intensity activity can be achieved through several exercise modalities. Although each type of exercise elicits distinct metabolic responses, all of them can improve glucose management; however, all of them also require careful prescription, monitoring, and personalization to reduce any diabetes-related risk factors and complications (28). This extends into ensuring programs are culturally inclusive and encompass a diverse profile of movement patterns and activity types, an area that needs more research to better elucidate impact. Recent reports suggest physical activity is highest amongst non-Hispanic White individuals when compared to Hispanic, non-Hispanic Black, and non-Hispanic Asian individuals (41). Additionally, exercise modality should best match the person and their preference—some of which may be impacted by cultural norms and religious background (42). Considering the expansive health benefits associated with recommended physical activity levels and their impact on glycemic control, this research must continue to extend into underrepresented groups to better understand how to best tailor exercise

prescriptions in culturally inclusive manners. To date, few studies have attempted to define best practices pertaining to exercise prescriptions among ethnic minorities and gaps remains in part due to heterogeneity of study design & findings, lack of clarity as to which exercise modalities are most impactful, and guidance on how to most precisely leverage peer and/or community health worker support for delivery (43–45). More research is cardinal to fill in these gaps.

DSMES & barriers to culturally competent DSMES training for providers

Part of a patient-centered approach to DSMES includes education and appreciation for how diverse ethnic and racial populations may differ in their phenotypic health profiles as well as culturally-specific attitudes and health behaviors (46). One such example is body composition and visceral adiposity (47). Asian ancestry populations often have increased intra-abdominal fat distribution compared to non-Asian groups. This influences the onset of T2D at lower BMI and potentiates cardiovascular disease risk (48). Current clinical practice guidelines encourage earlier vigilance and assessment of diabetes risk at lower BMI cut points in East Asian and Southeast Asian populations; despite this, several challenges and barriers often impede diabetes providers' ability to effectively engage with diverse diabetes populations including language discordance, patient cultural beliefs and practices, and lack of provider training in cultural competency and humility (49). Brown et al (2017) studied how lifestyle behaviors differed amongst ethnically diverse women at high risk for gestational diabetes (46). Researchers polled 1463 pregnant women (74% identified as part of a minority ethnic/racial group and 46% reported being born outside the US) in the research study "Gestational Diabetes Effects on Moms Trial". Some, but not all, health behaviors were found to be associated with ethnicity and were found to be different between groups. For example, ethnic identity was associated with fruit and vegetable intake, fiber intake levels, walking distance, and overall physical activity levels. More specifically, Latina women were found to consume the highest amounts of fruits and vegetables, Filipina women consumed the highest amount of fiber, and non-Hispanic White women consumed the lowest total calories (46). These findings could serve as a starting point for providers to explore how cultural background may differentially influence diabetes-related behaviors and facilitate discussions around personalized care.

Baker et al. examined racial/ethnic differences in physical activity and found that lower leisure time physical activity (LTPA) among Black and Hispanic participants was explained by educational attainment and overall health status, and no racial/ethnic differences were seen in total physical activity (TPA) after accounting for work-related physical activity (WRPA) (50). These findings were replicated by Saffer et al. in a nationally representative sample where they additionally demonstrated an inverse correlation between work-related and non-work physical activity with the latter being associated with greater health benefits (51). This suggests that

physical activity assessments during clinical encounters need to account for differences in work-related and leisure time demands while providing adequate counseling on their relative benefits when engaging patients around DSMES. Additional racial/ethnic nuances that one must account for include differences in body weight perception, experiences of weight-related stigma, and family roles and dynamics particularly as they relate to gender (52–54). Understanding how racial/ethnic and cultural differences may impact phenotypic profiles, perceptions, and health behaviors can assist providers in prescribing more patient-centered, relatable, and sustainable recommendations which can hopefully lead to the highest positive impact on diabetes management. Additional research and education in these areas are essential.

DSMES & health disparities: Medication & technology access and provider-patient communication

Diabetes care providers cover a wide spectrum of disciplines including dietitians, certified diabetes educators, physicians, advanced practice providers, endocrinologists, exercise physiologists, social workers, pharmacists, and both trained and untrained lay personnel such as community health workers and caregivers. As an individual diabetes patient journeys to incorporate DSMES into their lives, each of these providers may serve to enhance or detract from the success of T2D management. Medication adherence, access to diabetes technology, and the quality of patient-provider communication are additional critical components of DSMES success and are particularly sensitive to the aforementioned provider and patient-centered care factors. Diabetes patients from racial/ethnic minority groups have poorer medication adherence practices compared to non-Hispanic Whites with education and income levels, comorbidity burden, regimen complexity, psychosocial factors, and health beliefs often serving as important moderators of these associations (55–58). Additionally, cost, insurance type, health literacy status, and the digital divide can influence access to diabetes-related technologies, particularly for patients of color (59). Historically, the diversity of the trained diabetes workforce has been outpaced by the growing diversity within the US population, reflective of the broader healthcare workforce diversity needs (60). Several benefits of a more diverse diabetes workforce have been identified (61) including improved access to care for racial/ethnic minorities (62,63), greater engendering of trust and collaboration with diabetes patients including acceptance and application of DSMES recommendations, improved medication adherence (64,65), and greater satisfaction with care (66). This is relevant as several studies to date have shown that diabetes patients of color often report poorer communication with their providers compared to non-Hispanic White (NHW) patients (64), higher levels of medical mistrust (67–69), and greater experiences of diabetes distress and racial discrimination (70).

Efforts to increase workforce diversity, therefore, must be coupled with active commitments by educators to implement curricular innovations and enhancements aimed at training

and preparing diabetes professionals to enter a diverse practice arena. For example, exposure to anti-racism and implicit bias training, encounter role-playing with standardized diabetes patients, and effective health communication modules could aid in the acquisition of cultural humility skills among medical trainees toward these goals. Studies show that learner characteristics and experiences in the training environment (e.g., role modeling, interracial contact, racism, etc.) influence future healthcare providers' attitudes, preparedness, and willingness to provide high-quality care in several important diabetes-related constructs such as obesity management, empathic tendencies, and commitment to working with underserved populations (71–74).

Unfortunately, many practitioners in the current care environment may be ill-equipped to deliver culturally informed, evidence-based DSMES recommendations for diabetes patients from diverse backgrounds despite efforts of many training programs across the diabetes care spectrum to incorporate cultural competency training into their curriculums (75–77). Additionally, many patients of color continue to report a lack of access to culturally competent providers (78) and education support and materials (79). This likely reflects the lag time from training to practice but also highlights opportunities for educators to improve the standardization, content, implementation, and particularly their influence upon the continuing education of providers currently in practice.

Cultural competency relates to the ability to operate effectively in a variety of cross-cultural contexts and incorporates attitudes, beliefs, behaviors, communication, and policies that impact cross-cultural interactions in healthcare (80). This may be achieved by individual professionals, groups, institutions, and organizations, and manifests in specific practices, service delivery, support mechanisms, and outcomes when working with diverse populations (49). Cultural humility and sensitivity have emerged as complementary concepts to cultural competence; they refer to a dispositional posture of curiosity, self-reflection, and acceptance of potential biases, and an acknowledgment of the complexities of individual identities even within groups having shared cultural experiences. It recognizes that cultural competence, rather than being viewed as an endpoint, should inspire one to remain open and malleable to growth opportunities and mutual learning during and beyond the clinical encounter. Both concepts working in tandem best position diabetes providers to effectively engage in the shared decision-making and communication tasks required to achieve value-based and evidence-based action in diabetes care (81). We agree that both cultural competency and cultural humility/sensitivity are needed to engage in the challenging work of assisting patients from all racial/ethnic backgrounds to understand and apply the tenants of DSMES; however, challenges exist in their implementation in the training environment including the unintended risks of further stereotyping and “othering” patients and lack of exposure among trainees to diverse patient populations to gain the requisite skills to deliver culturally sensitive care (82). Still there is a final group of factors that pose threats to effective engagement in lifestyle change with which diabetes providers must also contend.

DSMES & social determinants of health factors (Figure 2)

Contextual factors

Despite the millions impacted by diabetes, research demonstrates that racial and ethnic minorities in the United States bear a disproportionate burden of worse health outcomes (83). Social determinants of health (SDOH), are used to better understand the contributions of external factors that may lead to health inequities, impact all levels of health, and the distribution of health-related resources and services. Hill-Briggs et al. (84) observe: “The social determinants of health are the conditions in which people are born, grow, live, work, and age. These circumstances are shaped by the distribution of money, power, and resources at global, national, and local levels” (84). SDOH factors such as income, education, access to insurance, and food security in neighborhood design can provide a better understanding of how an individual may be impacted in their management of diabetes care (85). Thus, the social, economic, and political structures all play a role in influencing individual health and provide context to lifestyle characteristics and conditions that impact behaviors. Reasons that certain groups may experience social and economic disadvantages over others, as well as the accessibility and availability of resources, including knowledge and information about diabetes management, are all influenced by a patient's social determinants of health. Those with low socioeconomic status (SES) for example, may face financial barriers in purchasing diabetes medication essential for self-management, which can impact adherence and blood sugar control maintenance or enrollment in educational programs or interventions (86); restricted access to healthy food options based on neighborhood characteristics (87); and lower health literacy levels (88), making it challenging to comprehend complex medical information, follow self-management guidelines and navigation of the overall healthcare system to manage their diabetes effectively. Moreover, different cultures may have their own perspectives on health, and gaining an understanding of specific cultural beliefs (89), practices, and influences on diabetes as an illness can help providers in their approach to diabetes management.

During patient visits, providers can gain an understanding of specific SDOH factors that may be most prominent in a patient's everyday life, and how they are most affected by these factors, such as access to digital health resources, resources providing education and knowledge on self-management strategies, and strategies to improve health based on contextual history, such as family history of health behaviors, practices, and beliefs. It would be beneficial to integrate a screening form that addresses SDOH, such as the Social Needs Screening Tool developed by the American Academy of Family Physicians (90), as part of a Self-Management assessment to provide a more comprehensive overview of contextual factors impacting patients with T2D. Also, the Cultural Formation Interview (91), developed by the American Psychiatric Association, is a tool that helps providers to better understand the impact of culture on a

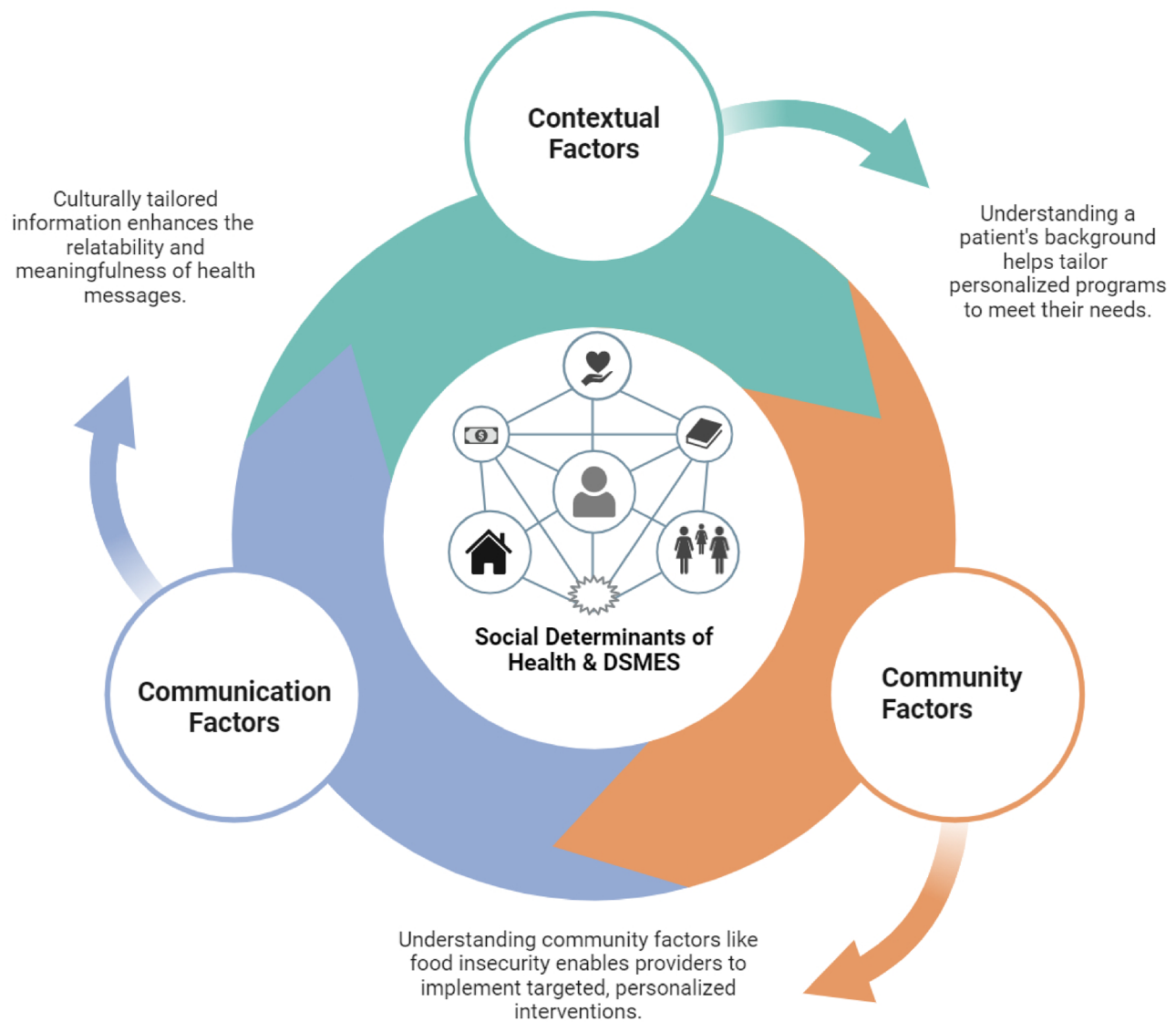


Figure 2. Social Determinants of Health & DSMES: External factors such as community, contextual, and communication aspects are critical for understanding and addressing health inequities. Recognizing these factors helps in tailoring personalized health interventions and improving the distribution of health resources and services, which is essential for achieving optimal diabetes care outcomes.

patient's health beliefs and behaviors by exploring cultural identity, perspectives on illness, stressors, and coping mechanisms. While it is used mainly for mental health, it can be adapted outside of mental health assessment and applied to other areas such as diabetes care. While cultural identity cannot be fully captured by a single tool, providers can use these cultural assessments along with open communication, dialogue, and empathetic listening to understand the context of patients' backgrounds and how to best approach their needs.

Community factors

Area-level factors in a community are a growing concern as contributors to the management of diabetes. The social and physical neighborhood characteristics in which an individual resides, particularly access and availability of nutritional resources, significantly impact diabetes self-management. Healthy food options can be severely limited in

neighborhoods with low socioeconomic conditions. Deprivation of adequate nutrition can lead to unhealthy food preferences and ineffective eating behaviors, which reduces energy levels and increases poor outcomes for individuals with chronic disease (92). The limited accessibility and availability of low-cost, healthy food options in safe and attainable ways is referred to as food insecurity (93). Food insecurity is increasingly prevalent among populations who are already at the highest risk for developing diabetes and diabetes-related complications (94). Factors associated with food insecurity include 1) higher stress and mental health issues (95), 2) limited access to educational programs or ineffective recommendations due to unavailable or inaccessible health food options (96), and 3) general difficulty in affording healthy foods and reliance on high-calorie and nutrient-poor options (97). Every one of these challenges negatively impacts blood sugar control.

Screening for potential food insecurity can lead to a better understanding and implementation of specific

interventions based on available resources and, in underrepresented communities, could help with glycemic control (98). One study found that incorporating nutritional education and food security factors helped patients improve HbA1c, particularly among low-income, predominately minority settings (99). Incorporating nutritional education based on community access to healthy foods, and other underlying factors can help tailor resources to what is available to the individual, and work toward developing specific management plans. As food insecurity significantly contributes to the likelihood of T2D, this is an issue that should be seriously considered. For example, the Indian Health Service Division of Diabetes Treatment and Prevention has developed a Food Insecurity Assessment Tool and Resource List which can be beneficial to help with strategies for patients to improve health outcomes and behaviors (Table 1).

Understanding area-level factors requires a multifaceted approach, including efforts to improve food access and affordability, promote nutrition education and skills development, and provide social support services. Collaborating with community organizations, implementing policies that address food insecurity, and integrating nutrition support into health-care systems and provider communications are important steps toward mitigating the impact of food insecurity on diabetes care management in culturally diverse populations.

Communication factors

Personalized health communication during patient visits with providers is also important to include in conversations about diabetes management. Nutrition education and dietary counseling as part of health visits for patients with chronic illness, are often inadequately addressed (100). Tailoring health information to culturally specific needs is particularly important. Cultural tailoring involves adapting educational materials, interventions, and strategies to align with the cultural beliefs, values, practices, and preferences of the community (101). Some ways this can be achieved could be by providing information in the native or preferred language (101), visual representations that resonate with the population (102), incorporating cultural dietary preferences (99), and recognizing variations in health literacy levels within culturally specific

populations (40). Furthermore, engaging families, community leaders, and support networks in the diabetes self-management education process, and training healthcare providers more on cultural sensitivity and awareness can make a significant community impact (103). By incorporating these cultural considerations, healthcare providers can enhance the relevance, accessibility, and effectiveness of diabetes self-management information for diverse cultural populations.

For example, one study examining responses from older African Americans found that the perceived quality of provider communication, including an understanding of culturally specific factors, significantly impacted their diabetes self-management efforts (104). Enhancing clinicians' cultural awareness to improve knowledge, skills, and practice that incorporates sociocultural characteristics of diverse groups can improve equitable care (40). Additionally, maintaining linguistically concordant communication among limited English proficiency populations is key to reducing the risk of misinterpretation, missing information, or limiting the chance to build rapport and patient satisfaction with provider interaction (105).

Appropriate communication can also influence the perception of risk and efficacy, and tailoring messages based on perceived risk can help construct adequate support plans for patients, who may differ in their beliefs of risk and degree of efficacy about diabetes (106). Furthermore, the availability of social support to manage lifestyle changes and health behavioral modifications to help improve diabetes should be considered (107). Social support and communication can vary among populations and ethnic groups, and while there is no one-size-fits-all approach, specific questions relating to the availability of this type of support can help gauge the extent of support, resources, or referrals needed (108). Quality communication should consider specific socio-cultural factors which can lead to an improved understanding of self-management behaviors, decision-making, and patient activation and empowerment (106). Publicly available resources can help provide an overview of cross-cultural diabetes educational strategies and an effective starting point for engaging in communication with different populations (Table 1).

Recognizing and addressing the unique cultural needs and contexts of individuals is vital for promoting optimal diabetes care outcomes. Culturally tailored information fosters a sense of cultural identity and respect, improving the relatability and meaningfulness of health to the population. This can assist with individuals being able to incorporate diabetes self-management strategies within their cultural frameworks, enhancing their ability to make informed decisions and adopt healthier practices. By taking a culturally specific approach to diabetes self-management education, healthcare providers can empower individuals from diverse cultures to take control of their diabetes and improve their overall health (103).

DSMES, cultural considerations, and practical guidance for next steps

Personalized and culturally tailored DSMES programs will require intentional next steps for successful adoption. Key tenets of personalized and culturally inclusive and tailored

Table 1. Publicly available resources providers can use to further their education or to share with their patients.

Resource Title	Access Link
<i>Food Insecurity Assessment Tool and Resource List</i>	https://www.ihs.gov/diabetes/education-materials-and-resources/index.cfm?module=productDetails&productID=332
<i>Cultural considerations: The role of diabetes care and education specialist A model for diabetes care</i>	https://www.diabeteseducator.org/docs/default-source/practice/educator-tools/cultural-considerations.pdf?sfvrsn=2
	https://archive.cdc.gov/#/details?q=model%20for%20diabetes%20care&start=0&rows=10&url=https://www.cdc.gov/vitalsigns/aian-diabetes/index.html
<i>Tasty recipes for people with diabetes and their families</i>	https://www.cdc.gov/diabetes/pdfs/managing/Tasty_Recipes_for_People_with_Diabetes-508.pdf

Food insecurity significantly contributes to the likelihood of T2D therefore, providers can and should share publicly available resources that can help patients with strategies to address and improve their diabetes management and overall health outcomes and behaviors.

diabetes care should include individualized diet programming and exercise prescription, updates to current provider training protocols and health communications systems, and programs in place to address and improve social determinants of health (Figure 3). One of the more successful examples of this approach includes community-based, participatory programs, which include community advisory boards equipped with key stakeholders to help improve communication between healthcare systems and the community, comprehensive programs that lean on local and accessible resources and programs, and providers who are trained to deliver individualized lifestyle care and management (109). It also includes systemic adjustments for how care is managed

from proper and mandatory provider training to local and policy-driven support to address any barriers related to the social determinants of health. An example of how to structure this approach includes a study by Nguyen *et al*, who leveraged an academic-community partnership to explore how a Community Engagement Continuum (CEC) model could be adapted to create a more culturally inclusive DMSES program tailored to Vietnamese Americans. The research team integrated a DSMES program into Vietnamese cultural practices by continuously incorporating community feedback to update the diabetes educational curriculum. The authors identified key topics that were important for the community, such as the need for increased nutrition

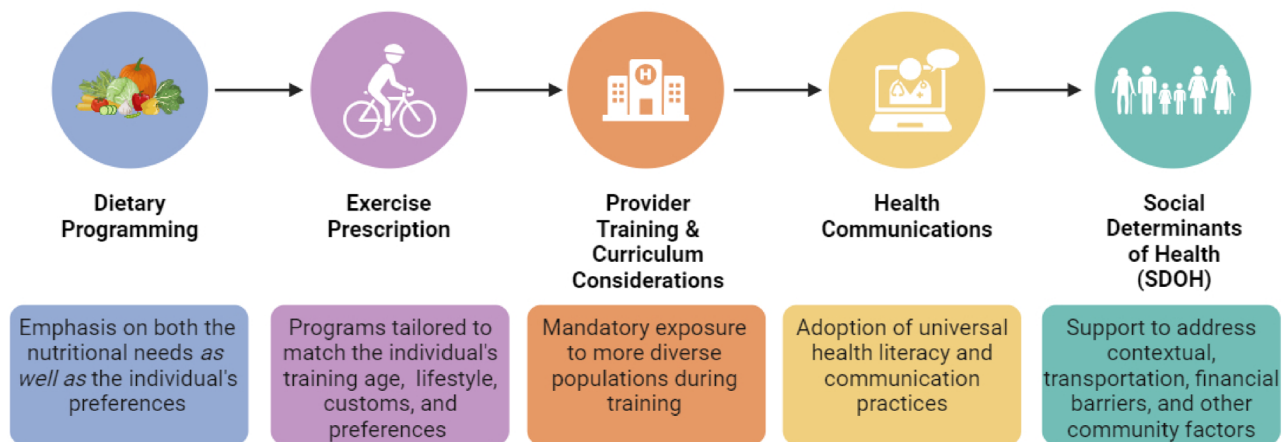


Figure 3. Key Tenets of Personalized and Culturally Inclusive Diabetes Care: Key tenets should include individualized diet programming and exercise prescription, updates to current provider training protocols and health communications systems, and programs in place to address and improve social determinants of health.

Table 2. Roadmap toward improved personalized diabetes care.

Factors Affecting DSMES	Proposed Approaches	Available Resources
Dietary Programming	Community engagement/feedback to best tailor food recommendations, recipes, meal programming, and more to individuals Knowledgebase of grocery stores, restaurants, and food options available and accessible in each community Personalized and culturally inclusive nutrition education programs	Focus groups, community family nights, and other social gatherings where nutrition feedback can be obtained Compiled list of available food locations with an updated menu and inventory list. This should be accompanied by a consistent review of the landscape to update the list in real time. Multi-lingual nutrition educational materials that incorporate culturally relevant food options/names and culturally inclusive visual aids for portion and food option recommendations
Exercise Prescription	Community engaged feedback to best tailor physical activity recommendations	Community events for physical activity programming with space for community feedback on preferred exercise modality, culturally relevant views on certain exercise programs, and preferred approach to increasing/maintaining weekly physical activity programming
Training & Curriculum Factors:	Exposure to more diverse populations as mandatory	Community-Academic partnerships; Inter-institutional collaborations between academic medical centers; learner-led clinics embedded in medical schools
Health Communications	Adoption of universal health literacy and health communication practices	Agency for Healthcare Research and Quality (ARHQ (https://www.ahrq.gov/))
SDOH	Adoption of universal health literacy and health communication practices Health systems incorporating community service support Leveraging additional Health Allied Staff Community engaged programs that encourage individuals to locate peer support for their social settings. Leveraging local social networks to help address diabetes self-management barriers Policy starting at the community level	Specific techniques such as: Teach back, Teach-to-goal Motivational interviewing Find Help Platform Social works, case managers, community health workers Open Table Platform (https://www.theopentable.org/)

The roadmap toward improved personalized diabetes care must include comprehensive, low-cost approaches that emphasize local, grassroots efforts through community engagement and a variety of accessible resources providers can share with individuals.

education, which had previously been a minimal focus for providers. The community input led to the team adjusting their DSMES approach to provide more relevant nutrition information, such as updating education materials to include Vietnamese food models, utensils, and food preparation methods (109). This study reinforces the importance of personalized and culturally tailored program and emphasizes the need for more resources and systems in place to aid in this pursuit. The roadmap toward improved personalized diabetes care must include comprehensive, low-cost approaches that emphasize local, grassroots efforts through community engagement and a variety of accessible resources providers can share with individuals (Table 2).

Conclusion

T2D is a growing epidemic that is disproportionately affecting underrepresented/minority communities throughout the US. Effective DSMES is the most impactful and cost-effective approach to prevent T2DM and is a critical component of managing T2D (7, 110). Cultural nuances do, and should, influence DSMES recommendations. A growing body of knowledge highlights the importance of including cultural considerations in diabetes management and extending this consideration for proper diabetes care and support. Providers should leverage a posture of cultural humility coupled with evidence based yet malleable DSMES that is delivered utilizing effective health communication skills to empower diabetes patients from varied backgrounds to succeed in diabetes care.

Future directions in this area are widespread but are imminent as prediabetes and T2D continue to rise, worldwide and throughout the US. There is a need for more translational research and application on how to best adapt DSMES to patients, and this work must include how approaches can be translated to suit a culturally diverse environment and improve patient outcomes. Health systems and providers should additionally be cognizant of the overgeneralization of cultural characteristics while accommodating patients' needs which can be approached through addressing individual preferences during encounters, understanding that cultural differences are not monolithic even among those who share a common cultural experience, and ensuring that diabetes communication remains transparent, empathetic, and consistent throughout the continuum of care.

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Ethics approval

Human and Animal Rights and Informed Consent: All reported studies/experiments with human or animal subjects performed by the authors have been previously published and complied with all

applicable ethical standards (i.e., Helsinki Declaration and its amendments, institutional/national research committee standards, and guidelines).

Author's contributions

CS, RW, and MS contributed drafting manuscripts. All authors contributed to concept, manuscript writing, critical evaluation, and article approval.

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Data availability statement

All reported studies/experiments have been previously published with available data to support findings.

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